ClipSAT : Cambridge & Edexcel Additional Mathematics 9606/4PM1 [ANSWER KEY]

Unit 4.01: Logarithmic & Exponential Functions

Q1. Solve the exponential equation 2^{x+2} = 5^x , giving x in terms of common logarithms.

[x] A: x = frac{2ln 2}{ln 5 - ln 2}
[ ] B: x = frac{2ln 5}{ln 5 - ln 2}
[ ] C: x = frac{ln 2}{2ln 5 - ln 2}
[ ] D: x = frac{2}{ln 5 - ln 2}

Solution: Take natural logs of both sides: x + 2ln 2 = xln 5 implies xln 2 + 2ln 2 = xln 5 implies xln 5 - ln 2 = 2ln 2 implies x = frac{2ln 2}{ln 5 - ln 2} . [M1][A1]

Q2. Solve the exponential equation 2^{x+3} = 5^x , giving x in terms of common logarithms.

[x] A: x = frac{3ln 2}{ln 5 - ln 2}
[ ] B: x = frac{3ln 5}{ln 5 - ln 2}
[ ] C: x = frac{3}{ln 5 - ln 2}
[ ] D: x = frac{ln 2}{3ln 5 - ln 2}

Solution: Take natural logs of both sides: x + 3ln 2 = xln 5 implies xln 2 + 3ln 2 = xln 5 implies xln 5 - ln 2 = 3ln 2 implies x = frac{3ln 2}{ln 5 - ln 2} . [M1][A1]

Q3. Solve the exponential equation 2^{x+4} = 5^x , giving x in terms of common logarithms.

[x] A: x = frac{4ln 2}{ln 5 - ln 2}
[ ] B: x = frac{ln 2}{4ln 5 - ln 2}
[ ] C: x = frac{4ln 5}{ln 5 - ln 2}
[ ] D: x = frac{4}{ln 5 - ln 2}

Solution: Take natural logs of both sides: x + 4ln 2 = xln 5 implies xln 2 + 4ln 2 = xln 5 implies xln 5 - ln 2 = 4ln 2 implies x = frac{4ln 2}{ln 5 - ln 2} . [M1][A1]

Q4. Solve the exponential equation 2^{x+5} = 5^x , giving x in terms of common logarithms.

[x] A: x = frac{5ln 2}{ln 5 - ln 2}
[ ] B: x = frac{5ln 5}{ln 5 - ln 2}
[ ] C: x = frac{5}{ln 5 - ln 2}
[ ] D: x = frac{ln 2}{5ln 5 - ln 2}

Solution: Take natural logs of both sides: x + 5ln 2 = xln 5 implies xln 2 + 5ln 2 = xln 5 implies xln 5 - ln 2 = 5ln 2 implies x = frac{5ln 2}{ln 5 - ln 2} . [M1][A1]

Q5. Solve the exponential equation 2^{x+6} = 5^x , giving x in terms of common logarithms.

[ ] A: x = frac{6ln 5}{ln 5 - ln 2}
[x] B: x = frac{6ln 2}{ln 5 - ln 2}
[ ] C: x = frac{6}{ln 5 - ln 2}
[ ] D: x = frac{ln 2}{6ln 5 - ln 2}

Solution: Take natural logs of both sides: x + 6ln 2 = xln 5 implies xln 2 + 6ln 2 = xln 5 implies xln 5 - ln 2 = 6ln 2 implies x = frac{6ln 2}{ln 5 - ln 2} . [M1][A1]

Q6. Solve the exponential equation 2^{x+7} = 5^x , giving x in terms of common logarithms.

[x] A: x = frac{7ln 2}{ln 5 - ln 2}
[ ] B: x = frac{ln 2}{7ln 5 - ln 2}
[ ] C: x = frac{7ln 5}{ln 5 - ln 2}
[ ] D: x = frac{7}{ln 5 - ln 2}

Solution: Take natural logs of both sides: x + 7ln 2 = xln 5 implies xln 2 + 7ln 2 = xln 5 implies xln 5 - ln 2 = 7ln 2 implies x = frac{7ln 2}{ln 5 - ln 2} . [M1][A1]

Unit 4.01: Logarithmic & Exponential Functions

Q7. Solve the exponential equation 2^{x+8} = 5^x , giving x in terms of common logarithms.

[x] A: x = frac{8ln 2}{ln 5 - ln 2}

[ ] B: x = frac{8ln 5}{ln 5 - ln 2}

[ ] C: x = frac{ln 2}{8ln 5 - ln 2}

[ ] D: x = frac{8}{ln 5 - ln 2}

Solution: Take natural logs of both sides: x + 8ln 2 = xln 5 implies xln 2 + 8ln 2 = xln 5 implies xln 5 - ln 2 = 8ln 2 implies x = frac{8ln 2}{ln 5 - ln 2} . [M1][A1]

Q8. Solve the simultaneous system: log_2x - log_2y = 2 and x - 3y = 3 . What is the value of x ?

[x] A: 12

[ ] B: 3

[ ] C: 6

[ ] D: 14

Solution: From the first equation: log_2x/y = 2 implies x/y = 2^2 = 4 implies x = 4y . Substitute into second: 4y - 3y = 3 implies y = 3 . Thus x = 43 = 12 . [M1][A1]